Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

V SEMESTER B. TECH (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, DECEMBER 2024

MODERN POWER CONVERTER [ELE 3124]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: DECEMBER 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Make a technical comparison between AT and ATX Power supply. ***(02)***

1B. State and prove ampere-second balance and volt-second balance. Also, establish the relationship between input voltage and output voltage for Boost-type DC-DC converters by applying ampere-second balance to the capacitor. ***(04)***

1C. Power supplies for telecommunications applications may require high currents at low voltages. Design a buck converter that has an input voltage of 6.6 V and an output voltage of 2.4 V. The output current varies between 4 and 6 A. The output voltage ripple must not exceed 2 percent. Specify the inductor value such that the peak-to-peak variation in inductor current does not exceed 40 percent of the average value. Determine the required capacitor. ***(04)***

2A. You are assigned to design a power supply to television sets which consume 16.384W power with a voltage rating of 48 V power from the source and load is 10Ω. Select a forword type DC-DC converter suited for this application.(i) Find the average output voltage of the converter ii) Find maximum and minimum values of inductor currents (ii) Design the number of turns and capacitors for this converter. Assume the output voltage ripples is 5%, the magnetizing inductor L_m= 5mH, the duty ratio is 0.4, and the switching frequency is 35 kHz. N_1/N_2 =1.5. ***(04)***

2B. The current-fed converter has an input voltage of 60 V and supplies a load of 80-Watt, 100 V and a switch duty ratio of 0.65. Determine (i) transformer turns ratio ii) the average current in the inductor iii) Maximum voltage across the switch. ***(03)***

2C. Analyze the load current, diode current, and inductor voltage waveforms of a Flyback DC-DC converter. Derive an expression to estimate the output voltage. ***(03)***

3A. You are assigned to design a power electronic LED driver suitable for a Honda City car LED Bumper Light of 12W with a voltage rating of 6V power from the car battery supply of 9 V. Select SEPIC type dc-dc converter suited for this driver.i) Find Duty ratio and average value of inductor current L_1. ii) output voltage ripple in capacitor C_1, inductor L_1=L_2= 90 µH, Capacitance C_1=C_2= 80µF and the switching frequency is 100 kHz. Analyze the inductor current I_{L1}. ***(04)***

3B. How to reduce the switching stress in a switched-mode power supply? explain with the help of a dissipative snubber circuit and switch loci with snubber. ***(4)***

3C.

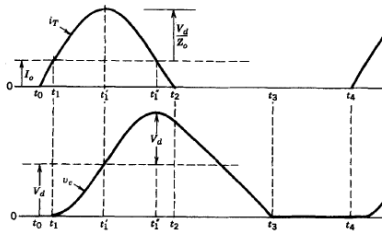


Fig.no.3C.

With the reference of Fig.Q.no 3C. identify suitable converter Topology in switched mode power converter. ***(02)***

4A. What are the drawbacks of Hard switching? Explain the possible solutions. ***(02)***

4B. Series Loaded Resonant Converter is having the source voltage of 150V, resonant inductor and capacitor are 23mH C_r= 0.159 µF, load resistance is R_L = 20Ω, and switching frequency is f_s = 200 kHz. Assume ω_s/ω_0 = 1.2, approximate quality factor is 2.9. Determine the output voltage of the converter. Assume the output filter components L_o and C_o produce a ripple-free output current and voltage. Justify the output voltage of the converter. Refer Fig.Q.no 4B.

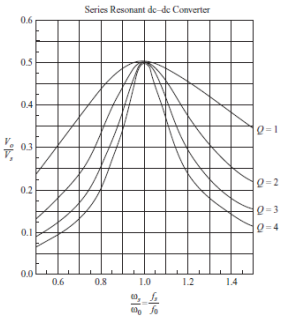


Fig.Q.no.4B.

(04)

4C. Analyze the waveform of the resonant inductor current and capacitor voltage of a buck converter. The switch is required to conduct a peak voltage that is higher than the source voltage V_s by an amount I_0Z_0 and explain its circuit properties. ***(04)***

5A. Derive the steady-state model of a boost converter. ***(04)***

5B. Differentiate between small signal and large signal modeling. ***(02)***

5C. Derive a small signal model of a boost converter. ***(04)***